Telecom customer churn

Logistic regression analysis and retention decisions

Assignment: 7_10_26_task.txt | Analysis date: 08 October 2026

Executive findings

The logistic regression model achieved a test ROC AUC of **0.843**. At the default 0.50 threshold, accuracy was **80.3%**, but only **55.1%** of actual churners were found. A validation-selected threshold of **0.13** found **92.8%** of churners, at the cost of more false alarms.

Test outcome	Default 0.50	Retention 0.13
Churners correctly identified (TP)	206	347
Non-churners wrongly flagged (FP)	110	457
Churners missed (FN)	168	27
Non-churners correctly classified (TN)	925	578

Dataset choice and scope

The supplied Position_Salaries.csv contains 10 job positions with Level and Salary. It has no churn label or customer attributes and cannot support the assigned classification task. This analysis therefore uses the Telco Customer Churn dataset explicitly linked in the task. The salary file was preserved unchanged.

The downloaded dataset contains **7,043 customers**, 21 columns and **1,869 churners (26.5%)**. There are 19 candidate inputs after removing customerID and the Churn target. This is a public example dataset, not evidence of performance on the commissioning company's customers.

Decision implication

The lower threshold is a candidate for inexpensive outreach when missing churners is costly. It flags 804 customers in the test set, compared with 316 at the default threshold. Offer cost, customer value and campaign effectiveness must determine whether that workload is worthwhile.

Source: Kaggle: Telco Customer Churn, as linked in the assignment; downloaded 08 October 2026. The Google Drive link in the task was not accessible.

Data and modeling method

Customer attributes and preparation

Inputs include demographics, tenure, phone and internet services, security and support services, streaming subscriptions, contract type, billing method and charges. Contract, tenure and charges are plausible churn predictors. Associations do not establish causes.

Customer IDs are unique and no duplicate rows were found. TotalCharges contained **11 blank values**, exposed by numeric conversion. Missing numeric values were imputed with training medians. Numeric features were standardized; categorical features were mode-imputed and one-hot encoded with a dropped reference category. Charge units are as supplied; no currency is assumed.

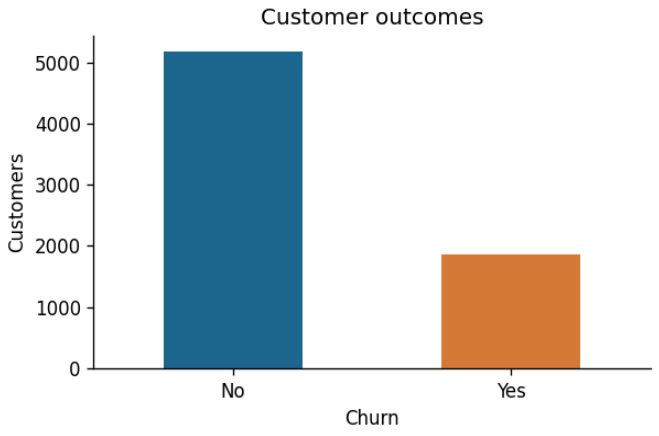
Split	Customers	Churners	Churn rate
Training	4225	1121	26.5%
Validation	1409	374	26.5%
Test	1409	374	26.5%

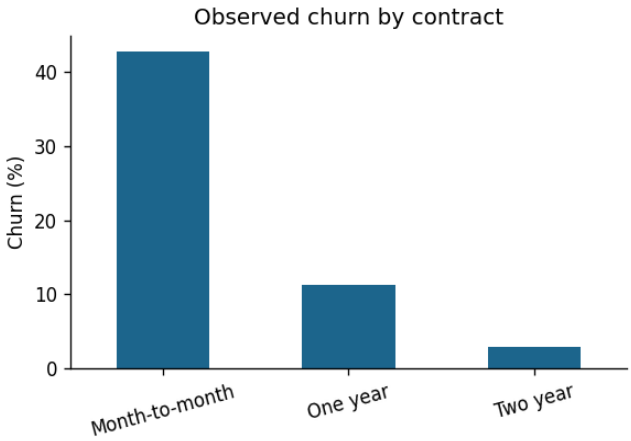
Splits were stratified with random seed 42: 60% training, 20% validation and 20% test. All learned preprocessing was fitted on training data inside the pipeline. IDs were excluded. Test outcomes were used only for final evaluation, not fitting or threshold selection.

Probability and classification

Logistic regression converts a weighted feature score z into a probability using P(churn) = 1 / (1 + exp(-z)). The model uses regularization strength C=1.0, the lbfgs solver and a maximum of 2,000 iterations. No resampling or class weighting was applied.

The default classification cutoff is 0.50. A second cutoff was selected from 0.10 to 0.80 in steps of 0.01 by maximizing validation F2, with larger cutoffs breaking ties. The selected cutoff was **0.13** (validation F2: 0.737). F2 prioritizes recall; it is not an economic cost function. The trained model was not refitted after selection.



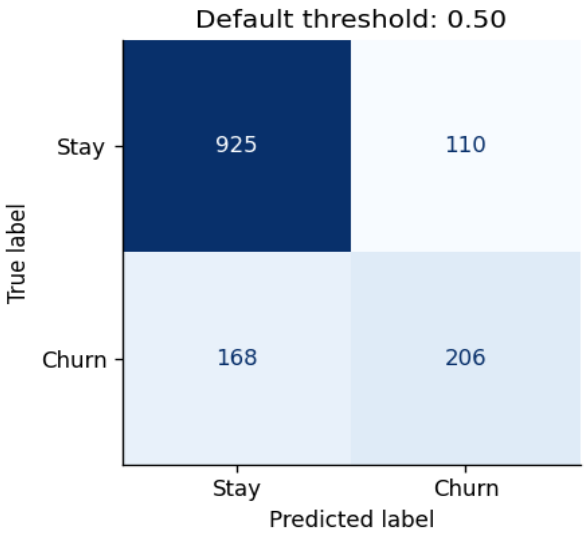


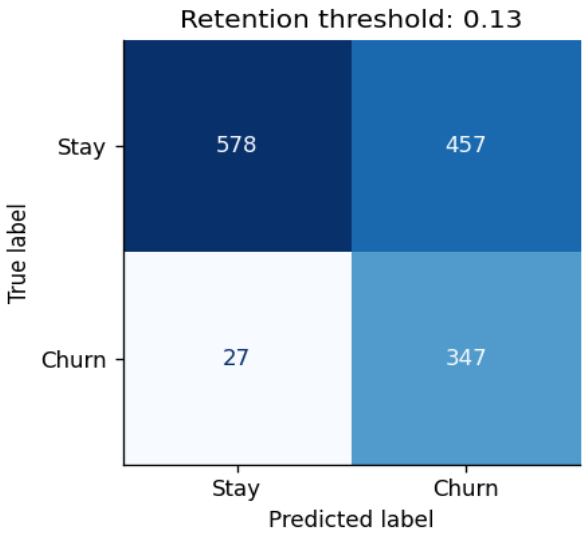
Performance on unseen customers

The held-out test set contains 1,409 customers: 374 churners and 1,035 non-churners. The same fitted probabilities support both thresholds.

Metric	Always stay	Default	Retention
Accuracy	0.735	0.803	0.656
Precision	0.000	0.652	0.432
Recall	0.000	0.551	0.928
F1	0.000	0.597	0.589
F2	0.000	0.568	0.754
ROC AUC	0.500	0.843	0.843
Average precision	0.265	0.635	0.635
Brier	0.195	0.138	0.138

The baseline always predicts stay and assigns the training churn rate as its constant probability. It achieves 73.5% accuracy while finding no churners. ROC AUC and average precision measure ranking; Brier score measures probability error (lower is better). Identical ranking scores across thresholds are expected.





Rows show actual outcomes; columns show predictions. The retention threshold finds **141 additional churners** but creates **347 additional false alarms**. Of its 804 flagged customers, 347 actually churned (43.2% precision). It misses 27 churners, versus 168 at the default cutoff.

Predictions and business interpretation

Examples from the unseen test set

These records were not used for training or threshold selection. Probabilities use the fitted model; labels use the selected 0.13 cutoff. The notebook exports predictions for all 1,409 held-out customers.

Customer ID	Probability	Prediction	Actual churn
4376-KFVRS	4.0%	Likely to stay	No
2754-SDJRD	70.2%	Likely to churn	No
9917-KWRBE	6.1%	Likely to stay	No
0365-GXEZS	38.0%	Likely to churn	No
9385-NXKDA	1.8%	Likely to stay	No
4686-UXDML	61.6%	Likely to churn	No

Which error is worse?

A false negative misses a customer who leaves, potentially losing future revenue and requiring acquisition spending. A false positive sends an unnecessary offer to a loyal customer. Missing a churner can be worse when customer value is high and outreach is inexpensive. False positives become more costly when discounts are substantial or contact capacity is limited.

An illustrative comparison is FP x unnecessary-offer cost + FN x missed-churn cost. A real campaign must also include offer costs to true positives and the probability that treatment prevents churn. No cost data or treatment-effect data were supplied, so the selected cutoff cannot be called profit-optimal. Detecting churn is not the same as preventing it.

One improvement to reliability

Collect recent, time-stamped customer data and validate on a later time period. This tests prediction for future customers more directly than a random split. Fit and tune on development data, then preserve a fresh final test period for an unbiased assessment.

Limits and reproducibility

Results describe one split of a public dataset. No confidence intervals, temporal validation or subgroup audit were performed. Correlated features and regularization affect coefficient interpretation. Gender and senior status are included for the assignment; deployment would require review of their suitability and subgroup performance. Brier score alone does not prove that probabilities are calibrated.

Customer_Churn_Analysis.ipynb includes data checks, fitted preprocessing, model training, threshold selection, plots, feature associations and a reusable prediction function. All code cells were executed successfully. The local dataset makes reruns independent of the download service. Source files remain unchanged.

Supporting outputs: evaluation_metrics.csv and unseen_customer_predictions.csv in output/. The notebook records package versions and the dataset SHA256 fingerprint.